EC991 Topics in Macroeconomics

Week 3 · The Aiyagari Model

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Where we stopped: the sequential problem

- A continuum of households of unit mass; future utility discounted by $\beta \in (0,1)$
- Earnings are $w\,y_t$: the market wage times an idiosyncratic **labour endowment** y_t, a Markov chain with transition probabilities $P(y' \mid y) = \Pr\left(y_{t+1} = y' \,\middle|\, y_t = y\right)$
- The only asset is a **riskless bond** paying r, with $R = 1 + r$, subject to the borrowing constraint $a_{t+1} \geq -\phi$
- The household's problem, given prices (w, r):

$$\max_{\{c_t,\, a_{t+1}\}} \; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + a_{t+1} = w\,y_t + R\,a_t, \qquad a_{t+1} \geq -\phi, \quad c_t \geq 0,$$

for $t = 0, 1, \ldots,$ with a_0 and y_0 given

- Prices carry no time subscript because we look for a **stationary** equilibrium: w and r are constant

The recursive formulation

- Everything the household needs to know today is its **state** (a, y): assets brought in and current endowment
- The same problem, written recursively (week 2's Bellman equation):

$$V(a, y) \;=\; \max_{a' \geq -\phi} \left\{ u\big(Ra + w\,y - a'\big) \;+\; \beta \sum_{y'} P(y' \mid y)\, V(a', y') \right\}$$

- The solution is a pair of **policy functions**: savings $a' = g_a(a, y)$ and consumption $c = g_c(a, y) = Ra + wy - g_a(a, y)$
- Two properties from week 2 that matter today:
  - the constraint binds at low wealth and low income: there, $g_a(a, y) = -\phi$
  - **precautionary saving**: at any given r the household saves more than it would under certainty

From partial to general equilibrium

- So far this is a **partial equilibrium** exercise: one household, prices fixed
- In **general equilibrium** the interest rate is the price that makes households' savings equal what the economy needs: r must clear the asset market
- We look for a **stationary equilibrium**: aggregates (total assets, output) and prices are constant through time, while every individual household keeps moving up and down the earnings and wealth distribution
- Such an equilibrium has a **stationary distribution** of households over (a, y): a distribution that replicates itself period after period
- Today: the stationary distribution and asset supply; closing the model with firms; existence; what the model delivers; then the first steps of computing it

Where households are: the distribution

- A household is characterised by its pair (a, y): assets and endowment
- The **distribution** at date t records what share of households sits at each pair:

$$\lambda_t(a, y) \;=\; \Pr\left(a_t = a,\; y_t = y\right)$$

- **Running example**: 100 workers, two asset levels $a \in \{1, 2\}$ and two endowments $y \in \{1, 2\}$, with 1 low and 2 high. Shares at date t, workers in brackets:

date t

	$y = 1$	$y = 2$
$a = 1$	0.40 (40)	0.20 (20)
$a = 2$	0.15 (15)	0.25 (25)

$\Longrightarrow$

date $t + 1$

	$y = 1$	$y = 2$
$a = 1$	?	?
$a = 2$	?	?

- The table **is** λ_t: the highlighted cell says $\lambda_t(1, 1) = \Pr(a_t = 1,\, y_t = 1) = 0.40$, i.e. 40 of the 100 workers
- **The question**: given where households are today, where are they tomorrow?

Where can one household go next?

- A household at (a, y) moves through two separate channels:
  - its **savings choice** fixes tomorrow's assets: $a' = g_a(a, y)$. No randomness here
  - the **endowment chain** draws tomorrow's income: y' with probability $P(y' \mid y)$

- The probability of reaching **one particular destination** (a', y') is the product of the two:

$$\underbrace{Q\big((a,y),\ \{(a',y')\}\big)}_{\text{probability of reaching this destination}} = \underbrace{\mathcal{I}\{g_a(a,y) = a'\}}_{\text{does the asset choice match? 1 if yes, 0 if no}} \times \underbrace{P(y' \mid y)}_{\text{probability of this endowment}}$$

- **Example**
  - Assume the savings policy: save when income is high, $g_a(a, 2) = 2$; dissave when it is low, $g_a(a, 1) = 1$
  - Assume the endowment chain: $P(2 \mid 1) = P(1 \mid 2) = 0.2$
  - Take a worker at $(1, 2)$, poor with high income. It saves, $a' = g_a(1, 2) = 2$, and then draws its income:

$$(1,2) \xrightarrow{a'=2} \begin{cases} (2,1) & \text{with probability } P(1 \mid 2) = 0.2 \\ (2,2) & \text{with probability } P(2 \mid 2) = 0.8 \end{cases}$$

  - It reaches $(1, 1)$ and $(1, 2)$ with probability 0: the asset choice does not match

Transitions

- For any group of destinations $\mathcal{A} \times \mathcal{Y}$: check whether the chosen assets belong to the group, then add up the probabilities of the endowments in it. That is the **transition function**:

$$Q\big((a, y),\, \mathcal{A} \times \mathcal{Y}\big) \;=\; \mathcal{I}\{g_a(a, y) \in \mathcal{A}\} \sum_{y' \in \mathcal{Y}} P(y' \mid y)$$

- **Example**
  - Assets $A = \{1, 2\}$ and endowment $Y = \{1, 2\}$, so tomorrow the household is in one of **four** states: $A \times Y = \big\{(1,1),\ (1,2),\ (2,1),\ (2,2)\big\}$
  - A household at $(1, 1)$ has low income, so it dissaves, $g_a(1, 1) = 1$. For each of the four states:

destination (a', y')	$(1,1)$	$(1,2)$	$(2,1)$	$(2,2)$
$\mathcal{I}\{g_a(1,1) = a'\}$	1	1	0	0
$P(y' \mid 1)$	0.8	0.2	0.8	0.2
$Q\big((1,1), (a', y')\big)$	**0.8**	**0.2**	**0**	**0**

  - So $Q\big((1,1), \cdot\big)$ gives the **probability of each of the four states** the household can be in tomorrow; the four numbers sum to one

Stationary distribution

- To get next period's distribution, apply the transition function Q to every point of today's distribution:

$$\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y}) = \int_{A \times Y} \underbrace{Q((a,y),\, \mathcal{A} \times \mathcal{Y})}_{\text{probability of reaching the destination set}} \underbrace{d\Lambda_t(a,y)}_{\text{population weight at the starting state}}$$

- This is the **law of total probability** applied to the population: tomorrow's share in a destination set is the sum of arrivals from every possible starting state

$A \times Y$	the **entire state space**, over which today's households are distributed
$\mathcal{A} \times \mathcal{Y}$	the **destination set** whose population share tomorrow we want
$\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y})$	the fraction of households in that destination set tomorrow

- We hold the **destination set fixed** and integrate over **all starting states**. $d\Lambda_t$ means "weight by today's distribution"; it is not a change in the distribution over time
- The **stationary distribution** is the one that replicates itself: for all (a, y), $\lambda(a,y) = \lambda_t(a,y) = \lambda_{t+1}(a,y)$
  - households still move between states; at every state, arrivals and departures balance

The law of motion in the example

- With four states, the population weight at a starting state is just its cell in today's table: $d\Lambda_t(a, y)$ becomes $\lambda_t(a, y)$, and the integral becomes a **sum over the four cells**:

$$\underbrace{\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y})}_{\text{tomorrow}} = \sum_{a \in \{1,2\}} \sum_{y \in \{1,2\}} Q\big((a, y),\, \mathcal{A} \times \mathcal{Y}\big)\, \underbrace{\lambda_t(a, y)}_{\text{today}}$$

- $\mathcal{A} \times \mathcal{Y}$ is a **generic destination set**. We now choose it in two different ways:
- **One cell**: $\mathcal{A} = \{1\}$, $\mathcal{Y} = \{1\}$, so $\Lambda_{t+1}\big(\{1\} \times \{1\}\big) = \lambda_{t+1}(1, 1)$:

$$\lambda_{t+1}(1,1) = \underbrace{0.8 \times 0.40}_{\text{from (1,1)}} + \underbrace{0 \times 0.20}_{\text{from (1,2)}} + \underbrace{0.8 \times 0.15}_{\text{from (2,1)}} + \underbrace{0 \times 0.25}_{\text{from (2,2)}} = \mathbf{0.44}$$

- **Two cells**: $\mathcal{A} = \{1\}$, $\mathcal{Y} = \{1, 2\}$ selects low assets and either income:

$$\begin{aligned} \Lambda_{t+1}\big(\{1\} \times \{1,2\}\big) &= \lambda_{t+1}(1,1) + \lambda_{t+1}(1,2) \\ &= 0.44 + \underbrace{0.2 \times (0.40 + 0.15)}_{\text{share reaching (1,2)}} = \mathbf{0.55} \end{aligned}$$

- So 55% have low assets tomorrow: we add the shares in the two destination cells

The law of motion in matrix form

- The matrix Q governs how the fraction in each state today is spread over the states tomorrow:

$$\underbrace{\begin{bmatrix} \lambda_{t+1}(a_1, y_1) \\ \lambda_{t+1}(a_1, y_2) \\ \lambda_{t+1}(a_2, y_1) \\ \lambda_{t+1}(a_2, y_2) \end{bmatrix}}_{\lambda_{t+1}} = \underbrace{\begin{bmatrix} Q_{11} & Q_{21} & Q_{31} & Q_{41} \\ Q_{12} & Q_{22} & Q_{32} & Q_{42} \\ Q_{13} & Q_{23} & Q_{33} & Q_{43} \\ Q_{14} & Q_{24} & Q_{34} & Q_{44} \end{bmatrix}}_{Q'} \underbrace{\begin{bmatrix} \lambda_t(a_1, y_1) \\ \lambda_t(a_1, y_2) \\ \lambda_t(a_2, y_1) \\ \lambda_t(a_2, y_2) \end{bmatrix}}_{\lambda_t}$$

  where Q_{mn} is the probability of moving from state m to state n, so $\lambda_{t+1} = Q'\lambda_t$ (rows of Q sum to one)

- Under certain conditions the Markov chain admits a **unique** stationary distribution $\lambda(a, y)$

Asset supply

- With the stationary distribution in hand, aggregate asset supply is the sum of the savings of all households
- That is, we integrate the savings policy against the distribution to get the **asset supply function**:

$$\mathbb{E}a(r) \;=\; \int_{A\times Y} g_a(a, y;\; r)\; d\Lambda(a, y;\; r)$$

- r enters twice: through the savings policy, and through the distribution itself
  - a change in r changes households' savings choices **and** the stationary distribution those choices generate
  - to obtain asset supply at another r, we must recompute **both**
- A higher interest rate increases the incentive to **postpone consumption**. Its effect on stationary aggregate asset holdings also depends on **income effects** and on changes in the **distribution**

Closing the model

To close the model, we must define the **demand for assets** in the economy. Two options:

- **Huggett (1993): credit economy.** Some agents borrow, others lend; the loan market clears when the aggregate demand for loans is zero:

$$\int_{A\times Y} g_a(a,y)\, d\Lambda \;=\; 0$$

- **Aiyagari (1994): production economy.** Firms demand capital to produce; the market clears when household savings equal capital demand:

$$\int_{A\times Y} g_a(a,y)\, d\Lambda \;=\; K$$

- We follow Aiyagari (1994) and assume an aggregate production function

Firms

- Production function: $Y = F(K, N) = K^{\alpha} N^{1-\alpha}$, with $\alpha \in (0, 1)$
- Capital depreciates at rate δ
- Markets are competitive; the firm's problem is standard (time subscripts omitted):

$$w = \frac{\partial F(K,N)}{\partial N} = (1-\alpha)\left(\frac{K}{N}\right)^{\alpha}, \qquad r + \delta = \frac{\partial F(K,N)}{\partial K} = \alpha\left(\frac{K}{N}\right)^{-(1-\alpha)}$$

- A **tight connection** between w and r through the capital-labour ratio: $\uparrow r \Leftrightarrow \downarrow K/N \Leftrightarrow \downarrow w$
  - a higher interest rate makes capital more expensive, so firms use **less capital per worker**
  - with less capital to work with, workers are less productive, and competitive **wages are lower**

Labour supply: the stationary endowment distribution

- There is a **continuum of households of mass one**, each supplying its endowment y as effective labour
- Endowments evolve according to the **exogenous** transition matrix P
- Let π_j be the stationary population share with endowment y_j. The vector π satisfies $\pi = P'\pi$, $\sum_j \pi_j = 1$
- **Two distributions**: π describes endowments y; Λ describes joint states (a, y)
  - in stationary equilibrium they agree on endowment shares: $\pi_j = \Lambda\big(A \times \{y_j\}\big)$, counting households with endowment y_j **across all asset holdings**
- Since labour supplied depends only on y, aggregate effective labour is

$$N = \sum_j y_j \pi_j$$

- **Example**: with the symmetric P, half have $y = 1$ and half have $y = 2$, so $N = 1 \times 0.5 + 2 \times 0.5 = \mathbf{1.5}$
- When we vary r, P and its unique stationary distribution π remain fixed. Hence N **remains fixed**, although savings choices and the joint distribution change

Equilibrium definition

A **stationary recursive competitive equilibrium** is a value function V; household policy functions g_a and g_c; firm choices K and N; prices w and r; and a stationary distribution Λ such that:

1. Given prices, V, g_a and g_c solve the **household problem**
2. Given prices, K and N solve the **firm's problem**
3. Given the transition function Q, the stationary distribution satisfies $\Lambda(\mathcal{A} \times \mathcal{Y}) = \int Q\big((a, y), \mathcal{A} \times \mathcal{Y}\big)\, d\Lambda$
4. The **labour market** clears: $N = \sum_j y_j \pi_j$
5. The **asset market** clears: $\int g_a(a, y)\, d\Lambda = K$
6. The **goods market** clears: $\int g_c(a, y)\, d\Lambda + \delta K = F(K, N)$

Equilibrium in one unknown: the interest rate

The test: for each candidate interest rate r, compare the capital firms demand with the assets households supply

$$Z(r) \;\equiv\; K(r) - \mathbb{E}a(r)$$

- $Z(r) > 0$: firms want more capital than households supply
- $Z(r) < 0$: households supply more assets than firms want to use
- $Z(r) = 0$: the plans match, and the asset market clears. **An equilibrium is an r^* with $Z(r^*) = 0$**

- **One price is enough**: N is fixed by the endowment process. Given N, each r fixes the firm's capital demand, and the resulting K/N fixes the wage:

$$K(r) \;=\; N\left(\frac{\alpha}{r+\delta}\right)^{\frac{1}{1-\alpha}}, \qquad w(r) \;=\; (1-\alpha)\left(\frac{K(r)}{N}\right)^{\alpha}$$

- **Asset supply at each r**: solve the household problem at **both** prices $\big(r, w(r)\big)$, find the stationary distribution Λ, and aggregate to $\mathbb{E}a(r)$

- **To show an r^* exists**: Z continuous, with a **sign change** between two interest rates

Households' desired wealth

Households' supply: we have firms' demand $K(r)$. Now we need asset supply $\mathbb{E}a(r)$, particularly near the highest sustainable return

- **Income risk encourages saving**: wealth protects consumption against bad income shocks
- **Protection has diminishing benefits**: another unit of wealth matters less when the buffer is already large
- **Impatience encourages consumption today**: when $\beta R < 1$, the return does not fully compensate for waiting
- **A finite buffer balances these forces**: eventually, the extra protection from one more unit of wealth is too small to justify postponing more consumption
- **As r rises**, waiting is better rewarded, so households are willing to hold **larger buffers**
- **As** $\beta R \to 1$: ever larger buffers are needed before the extra protection is small enough to stop saving

Under the standard Aiyagari assumptions:

$$r \to \bar{r} \equiv \frac{1}{\beta} - 1 \qquad \Longrightarrow \qquad \mathbb{E}a(r) \to +\infty$$

Capital demand $K(r)$ remains finite, so near this boundary there is **excess asset supply**:

$$Z(r) = K(r) - \mathbb{E}a(r) < 0 \qquad \text{for } r \text{ sufficiently close to } \bar{r}$$

Finding an interest rate that clears the asset market

- **At sufficiently low interest rates**: capital is cheap for firms, while saving offers households a low return. Firms demand more capital than households supply:

$$Z(r_L) \;=\; K(r_L) - \mathbb{E}a(r_L) \;>\; 0$$

- **Near the upper boundary**: from the previous slide, household asset supply exceeds firms' capital demand:

$$Z(r_H) \;=\; K(r_H) - \mathbb{E}a(r_H) \;<\; 0$$

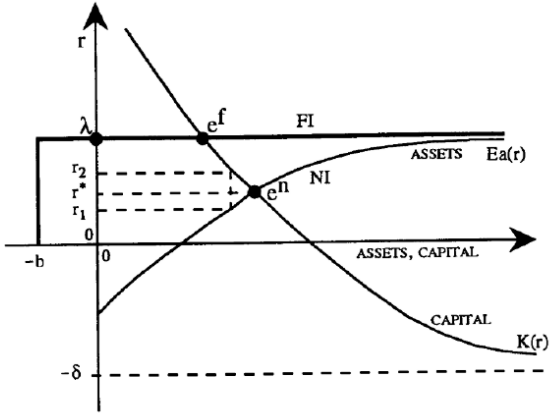
- **If excess demand changes continuously**: somewhere between these rates, household savings and firms' capital demand match:

$$K(r^*) \;=\; \mathbb{E}a(r^*)$$

- At least **one** equilibrium exists; there may be more than one

More details

General equilibrium: where supply meets demand



Aiyagari 1994, Figure IIb. Illustrative curve shapes; uniqueness is not guaranteed.

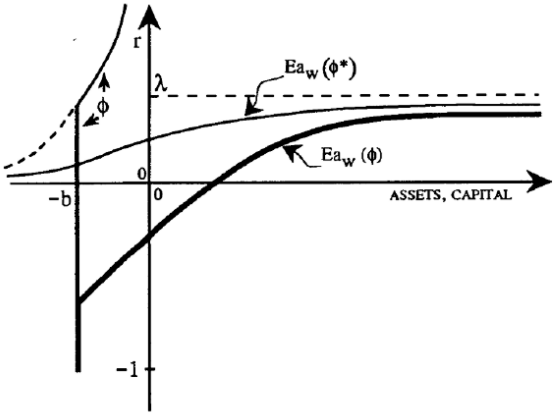
Curves
$K(r)$: firms' capital demand.
$\mathbb{E}a(r)$: household asset supply.
NI: no insurance against income risk.

Incomplete-markets equilibrium
At e^n: $\mathbb{E}a(r^*) = K(r^*)$.
Below r^*, at r_1: excess demand.
Above r^*, at r_2: excess supply.

Full-insurance benchmark
FI: full insurance; equilibrium at e^f.
At e^n: more capital and a lower r.

Original notation
$\lambda = \bar{r} = 1/\beta - 1$: a rate,
not the household distribution.

Asset supply: the role of borrowing limits



Aiyagari 1994, Figure IIa. For $r > 0$, the natural debt limit is $\phi^* = wy_{\min}/r$.

Two borrowing restrictions
Thick curve, $\mathbb{E}a_w(\phi)$:
tighter borrowing limit.
Thin curve, $\mathbb{E}a_w(\phi^*)$:
borrowing up to the natural debt limit.

Read horizontally
In the comparison shown, more borrowing flexibility means fewer assets.

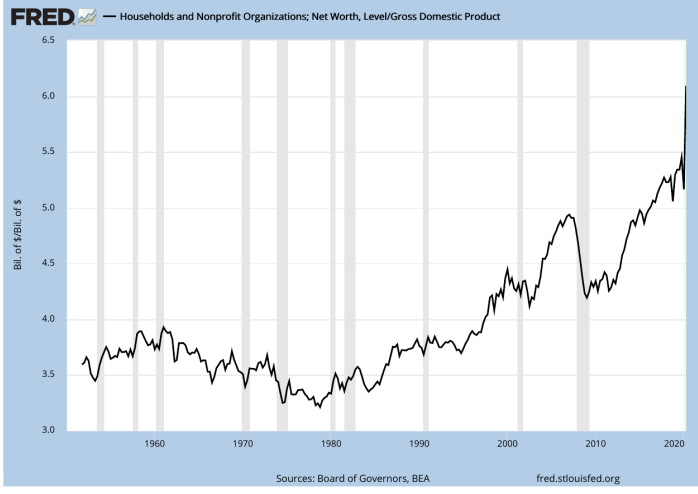
Intuition
Borrowing can help cover low income.
Households rely less on saved assets.

Original notation
w in $\mathbb{E}a_w$: wage held fixed.
$-b$: imposed asset floor (our $-\phi$).
$\lambda = \overline{r} = 1/\beta - 1$.

Calibration

- **Standard parameters**: $\alpha = 0.33$, CRRA utility with γ between **1 and 5**, $\delta = 0.08$
- **Discount factor** β: chosen to match the aggregate or average **wealth-to-income ratio**
  - option 1, **macro target**: aggregate household wealth over GDP from the national accounts; for the U.S. about **3 to 5**, depending on the period and on whether residential capital is included
  - option 2, **micro target**: average wealth from a household survey, such as the Survey of Consumer Finances (SCF)
- **Labour income process**: estimated from micro data on individual income dynamics, as an AR(1) $\log y_t = \rho \log y_{t-1} + \sigma \varepsilon_t$, then discretised to a Markov chain (this week's tutorial)
- **Borrowing limit** ϕ: chosen to match the fraction of households with **negative net worth**, around **10%** in the U.S. (or set $\phi = 0$)

Calibration target: aggregate wealth-to-GDP ratio, U.S.



- Household net worth relative to GDP stayed between about **3** and **4** until the mid-1990s and has been between about **4** and **5.5** since: the value of the target depends on the period

FRED: Households and Nonprofit Organizations; Net Worth, Level / Gross Domestic Product (Board of Governors, BEA)

Calibration target: average household wealth (SCF 2016)

Wealth definition		All	Excluding top 1%
Total	Mean	11.2	7.8
	Median	1.6	1.7
Non-housing	Mean	8.5	5.3
	Median	0.5	0.5
Financial	Mean	5.3	4.0
	Median	0.3	0.3
Mean earnings		$61,600	$54,900
Mean income		$102,200	$84,300

- Wealth is expressed as a ratio to mean household **earnings**
- Average wealth $= 11.2 \times \$61{,}600 = \$689{,}920$
- Average wealth / average income $= \$689{,}920/\$102{,}200 =$ **6.75**
  - surprisingly hard to square with the aggregate ratio on the previous slide

Survey of Consumer Finances 2016

Calibration target: low-wealth households (SCF 2016)

Wealth definition		Share of households
Total	≤ 0	11%
	$\leq$ \$2,000	17%
Non-housing	≤ 0	15%
	$\leq$ \$2,000	22%
Financial	≤ 0	11%
	$\leq$ \$2,000	31%

- About **11%** of households have zero or negative total net worth: the target for the borrowing limit ϕ
- Many more hold almost nothing: **31%** have \$2,000 or less in financial wealth

Survey of Consumer Finances 2016

Why does precautionary saving raise the saving rate only modestly?

Result (Aiyagari 1994): for reasonable calibrations, the difference between saving rates under complete and incomplete markets is **small**, at most **2%**

A force and a counteracting feedback

- **The force**: uninsured households want **larger asset buffers** to protect consumption against bad income shocks
- **The feedback**: the additional assets finance additional capital. More capital per worker lowers its marginal product, so the interest rate **falls**
- **The consequence**: the lower return discourages some of the extra accumulation households would have wanted at unchanged prices, so the **equilibrium** increase in saving is limited

When can the difference be larger?

- With much larger or more persistent income shocks, the precautionary motive is stronger, so the saving-rate difference can be **substantially larger**
  - **larger** σ: bad shocks are deeper, so households need a larger buffer
  - **larger** ρ: low income lasts longer, so savings must cover a longer spell

Aiyagari 1994, p. 672

Why wealth > income > consumption inequality

Result (Aiyagari 1994): inequality ranks as in the data: **wealth** > **income** > **consumption**

Households transfer resources across dates: this **cushions consumption** and carries **earnings histories** into wealth

Consumption responds less to temporary income differences

- Concave utility encourages households to spread consumption across dates
- They save part of favourable earnings and draw down assets or borrow after adverse shocks
- Borrowing constraints restrict smoothing; more persistent earnings shocks require larger consumption adjustments

Why wealth > income: asset differences outlast earnings differences

- Exogenous earnings histories differ; optimal saving decisions translate them into asset holdings
- Current earnings can become similar again while accumulated assets remain different
- Households with little wealth still earn wages: when earnings are less concentrated than assets, total income is less unequal than wealth

At the comparison date	Household A	Household B
Past earnings	Favourable history	Adverse history
Current earnings	Same	Same
Expected future labour earnings	Same	Same
Accumulated assets	**Larger buffer**	**Smaller buffer**

Wealth is right-skewed, but not concentrated enough

Result (Aiyagari 1994): the wealth distribution is **right-skewed**, so top inequality is a feature of these models

But the baseline is quantitatively off

- Wealth inequality in the model is **lower** than in the data
- In the SCF, mean wealth is **11.2** times mean earnings, while median wealth is only **1.6** times: a few households hold very large fortunes

Why: limited incentives to retain enormous fortunes

- Once wealth provides substantial protection against earnings risk, additional insurance is less valuable
- With $\beta(1+r) < 1$, impatience then encourages wealthy households to **consume their assets**

Conclusion

- The incomplete-markets model delivers **new theoretical insights** and reopens old questions: capital taxation, government debt, …
- But where the model **shines** is as a framework for new questions about **income and wealth inequality**
- A large subsequent literature works to make the model **match the wealth distribution** well
- And then studies policies where inequality is central: progressive taxation, social security, …

Where to go now?

- **Progressive taxation**: Boar and Midrigan (2022, JME); Kindermann and Krueger (2022, AEJ: Macro); Brüggemann (2021, AEJ: Macro); Heathcote and Tsujiyama (2021, JPE); Conesa, Kitao and Krueger (2009, AER)
- **Hours and labour market frictions**: Bick, Blandin and Rogerson (2022, QJE); Krusell, Mukoyama and Şahin (2010, ReStud)
- **Top wealth inequality and heterogeneous returns**: De Nardi and Fella (2017, RED); Benhabib, Bisin and Luo (2019, AER); Benhabib, Bisin and Zhu (2011, ECTA); Kaymak, Leung and Poschke (2022)
- **Liquid and illiquid assets, the wealthy hand-to-mouth, MPCs**: Kaplan and Violante (2014, ECTA; 2022, Annual Review)
- **Unsecured credit and consumer default**: Chatterjee et al. (2007, ECTA); Livshits et al. (2007, AER); Dempsey and Ionescu (2022); Herkenhoff (2019, ReStud)
- **Non-homothetic preferences**: Straub (2022); Mian, Straub and Sufi (2021, QJE); Carroll and Hur (2020, JME); Känzig (2022)

Extension: Fiscal Policy

- Households face uninsured earnings risk and use assets to protect consumption
- Fiscal policy changes households' resources, saving opportunities and equilibrium prices
- **Can these changes improve welfare after accounting for their financing and production costs?**

Introduce government debt

Households' financial assets now finance both productive capital and government debt:

$$\boxed{A = K + B}$$

- A: assets held by households
- K: productive capital
- B: government debt held by households
- **Household assets and productive capital are no longer the same object**

How taxes and transfers enter household decisions

The household budget becomes:

$$(1+\tau_c)\, c_t + a_{t+1} \;=\; (1-\tau_w)\, w y_t + \left[1 + r(1-\tau_r)\right] a_t + \tau$$

- **Labour-income tax** τ_w: households retain less of their earnings
- **Capital-income tax** τ_r: households retain less of the return to saving
- **Consumption tax** τ_c: consumption costs more
- **Transfer** τ: households receive additional resources

Intuition

- Transfers can support consumption when earnings or assets are low
- Financing those transfers changes disposable resources and potentially saving incentives
- Households choose consumption and saving taking the **entire tax-and-transfer system** into account

Government debt must be financed

The government pays for purchases and debt repayment using net taxes and new borrowing:

$$G_t + (1 + r)B_t \;=\; B_{t+1} + T_t^{\text{net}}$$

In steady state, debt is constant: new borrowing replaces maturing principal. Therefore,

$$\boxed{G + rB \;=\; \tau_w\, wN + \tau_r\, rA + \tau_c\, C - \tau}$$

- Taxes finance government purchases, interest payments and transfers
- Transfers subtract from net revenue
- **More debt creates an interest bill that must be financed**

Our policy experiment

- Increase B, keeping G, τ, τ_c, τ_r fixed
- Adjust τ_w so the government budget holds
- Because labour supply is fixed here, this tax adjustment changes disposable earnings, **not hours worked**

More debt: what happens to capital and returns?

Household asset holdings respond to the new policy. Market clearing requires:

$$K = A - B$$

- If household assets increase by less than public debt, **productive capital falls**
- With less capital per worker, capital's marginal product rises: $K/N \downarrow \implies r \uparrow,\ w \downarrow$
- Lower capital also means lower output

Intuition

- Some household wealth now finances government debt instead of productive capital
- **Capital can fall even if households hold more financial assets**

Government debt: better self-insurance at a cost

Potential benefit

- Precautionary saving raises capital, but households remain imperfectly insured
- Government debt provides another asset for smoothing consumption: $A = K + B$
- A higher return can make asset buffers more effective at supporting consumption

Costs

- When debt crowds out capital, wages and output fall
- Servicing debt requires taxes, changing households' disposable resources
- Welfare depends on these effects together, including who gains and who loses

Does the policy improve welfare? Weighing the two

The question: do improved self-insurance opportunities outweigh the losses from lower production and the distributional effects of financing?

- A welfare improvement under a chosen social criterion need not benefit every household
- Aiyagari–McGrattan find positive optimal debt and **small** gains from moving from existing debt to the optimum
- Life-cycle extensions can instead favour **public asset accumulation**

Aiyagari and McGrattan (1998, JME); Peterman and Sager (2022, AEJ: Macro)

Backup: excess demand at low interest rates

- As $r \to -\delta$, capital costs $r + \delta \to 0$ per unit, so $K(r) \to +\infty$
- This alone does not sign Z: the wage $w(r)$ also rises without bound, and richer households may save more
- So compare the two sides **in units of the wage**:

$$\frac{K(r)}{w(r)} = \frac{\alpha}{1-\alpha}\frac{N}{r+\delta} \to +\infty$$

- When $\beta R < 1$, households hold a finite buffer, so $\mathbb{E}a(r)/w(r)$ stays bounded under the maintained assumptions
- Hence $Z(r) > 0$ for r sufficiently close to $-\delta$

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